Parameter-map doc — revisions per Haochen's feedback

Parameter-map doc — revisions per Haochen’s feedback

This companion note summarises the changes made to `dnaa5_oric_saturation_parameter_space.md` in response to Haochen’s three comments on the first version, and flags what is still open.

Haochen’s comment 1 — scan shape parameters within a chosen form

> Part 3. Cooperativity functional form at oriC: when exploring different function forms, also explore parameters of those forms. For example, for Hill function, not only explore Hill coefficient n (how sharp the transition is), but also scan different K (where the transition happens).

Change made: §3 now ends with an explicit note that within a chosen form, the shape parameters must be scanned too — for Hill specifically, scan **both** the Hill coefficient `h` (sharpness) **and** `K` (midpoint).

Haochen’s comment 2 — hydrolysis at oriC should not be the cluster-reset mechanism

> Part 5. DnaA-ATP hydrolysis at oriC (cluster reset): I’m confused by this part, because cluster reset should always happen right after initiation because all bound DnaA at oriC should fall back to the cytoplasm at that moment. Because the time window of oriC saturation is typically much shorter than DnaA-ATP half-life, the hydrolysis of DnaA-ATP bound at oriC should be negligible and should not be the main reason for the cluster reset. Please double-check if this is true in the current implementation.

Code check: the comment in `dnaa_box_binding.py` explicitly says *"fork release is what actually destroys the state."* So fork release at initiation IS the dominant reset mechanism in the current implementation. Hydrolysis at oriC runs in the background at `k_h = 0.025/min` (uniform across all DnaA-ATP pools, bound and free), but it is not what cleans out the cluster after initiation.

Change made: §5 (the hydrolysis / cluster-reset section) has been removed entirely. It was framing hydrolysis as a tuning lever for cluster reset, which is incorrect. Sections downstream have been renumbered; the hydrolysis entry has also been dropped from the ranked-knobs list.

Haochen's comment 3 — try real initiation triggering by oriC saturation

> One thing I realize is that maybe we should do real initiation triggering at this stage — not artificially by mass, but by oriC saturation events. My concern is that initiation itself is a feedback to oriC state — if oriC saturation is too fast, initiation happens early, which will generate more DnaA boxes during replications and titrate redundant DnaA-ATP; if oriC saturation is too slow, initiation will not happen, allowing DnaA-ATP to keep accumulating, until oriC saturates. Without this feedback, you might always see over- or under-saturation at oriC, as you and Eran recently obtained, but it can be an artifact by mass-triggering initiations. The caveat is that this could mess up the regular cell cycles (being very long or short) and affect other processes in the background. I would still give it a try and see what would happen.

Status: open. Not yet incorporated as a documented parameter-space axis.

Next step proposed: replace the mass-clock gate in `chromosome_replication.py` (currently `criticalMassPerOriC ≥ 1.0`) with an oriC-saturation gate (init fires when at least one chromosome's oriC_low cluster is fully bound). This is a code change rather than a parameter sweep; to be tried as a focused experiment with the current best lineage config (V=1.5 + Hill K_d) to see whether the feedback Haochen describes does in fact clamp the late-gen drift.